Test report: DM maintenance scripts

Item	Value
Scripts	`run_maintenance.sh`, `tables/dm.files.sql`, `tables/dm.filerequests.sql`
Server	Local Docker test container (PostgreSQL 16.15 (Debian))
Database	`testdb`
Run	2026-10-24 13:18:25 to 2026-10-24 13:20:02 (server time)

13 of 13 scenarios passed

Before every scenario `tests/reset_data.sql` recreates the dummy data (schema `dm`), and any extra data the scenario needs is inserted. `run_maintenance.sh` is then run, answering only the connection questions, and the real result is checked with SQL queries against the tables. Rerun everything with `PGHOST=<server> PGUSER=postgres ./tests/run_tests.sh`.

Summary

ID	Scenario	Expected	Result
S01	Full run: partition N emptied, filerequests keeps statusid 1, 2	files_active ('N') emptied, files_removed ('Y') untouched; filerequests keeps 2,000 rows, all statusid 1 or 2; dated copy holds 2,000; exit code 0.	PASS
S02	Table named on the command line: only that one runs	dm.filerequests done (2,000 rows); dm.files never started (files_active still 2,000); exit code 0.	PASS
S03	Table without a script is refused, nothing runs	'ERROR: no script tables/dm.no_such_table.sql' before the connection prompt; nothing changed; exit code 1.	PASS
S04	Connection error is reported, nothing runs	'ERROR: cannot connect', nothing changed, exit code 1.	PASS
S05	Commented-out table is not touched	dm.files done; dm.filerequests never started and still 5,000 rows; 'COMPLETED: 1 table(s)'.	PASS
S06	Partition missing: run stops, next table not run	'relation "dm.files_active" does not exist', dm.files FAILED, dm.filerequests never started, exit code 1.	PASS
S07	Hard-coded partition: exactly the named partition is emptied	Exactly the named partition (files_removed, 3,000 'Y' rows) is emptied, with no check; files_active and files_other untouched; exit code 0.	PASS
S08	Foreign key: TRUNCATE fails, table rolled back	TRUNCATE refused, everything undone: 5,000 rows, no copy left, exit code 1.	PASS
S09	Keep condition matching 0 rows is refused	'Copy check failed', table unchanged, no copy left, exit code 1.	PASS
S10	Invalid keep condition is rolled back	Error 'column does not exist', table unchanged, no copy left, exit code 1.	PASS
S11	Second run on the same day fails safely	Second run stops: today's maintenance table already exists; table and copy keep 2,000 rows; exit code 1.	PASS
S12	No count(*) on the main table	Counter +3: 1 = the copy step, 2 = TRUNCATE rebuilding the 2 indexes on the empty table. No count(*) on the table.	PASS
S13	Table in use: lock timeout after 60 s	After 60 s 'canceling statement due to lock timeout'; nothing changed, no copy left, exit code 1.	PASS

Scenario details

S01. Full run: partition N emptied, filerequests keeps statusid 1, 2: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	./run_maintenance.sh (static list: dm.files, dm.filerequests)
Expected	files_active ('N') emptied, files_removed ('Y') untouched; filerequests keeps 2,000 rows, all statusid 1 or 2; dated copy holds 2,000; exit code 0.
Checked	files_active | files_removed | filerequests | wrong statusid | rows in copy | exit code
Expected values	`0|3000|2000|0|2000|0`
Actual values	`0|3000|2000|0|2000|0`
Exit code	0

```
2026-10-24 18:48:25 START DM maintenance on postgres@localhost:55432/testdb: dm.files dm.filerequests
2026-10-24 18:48:25 START  dm.files
dm.files done: partition dm.files_active emptied
2026-10-24 18:48:25 DONE   dm.files
2026-10-24 18:48:25 START  dm.filerequests
copy OK: 2000 rows
check OK: 2000 rows copied back
dm.filerequests done: kept 2000 rows, removed about 3000 rows, copy in dm.maintenance_filerequests_20261024
2026-10-24 18:48:25 DONE   dm.filerequests
2026-10-24 18:48:25 COMPLETED DM maintenance: 2 table(s). Log: logs/dm_maintenance_20261024_184825.log
```

S02. Table named on the command line: only that one runs: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	./run_maintenance.sh dm.filerequests
Expected	dm.filerequests done (2,000 rows); dm.files never started (files_active still 2,000); exit code 0.
Checked	filerequests | files_active | dm.files started | exit code
Expected values	`2000|2000|no|0`
Actual values	`2000|2000|no|0`
Exit code	0

```
2026-10-24 18:48:25 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:25 START  dm.filerequests
copy OK: 2000 rows
check OK: 2000 rows copied back
dm.filerequests done: kept 2000 rows, removed about 3000 rows, copy in dm.maintenance_filerequests_20261024
2026-10-24 18:48:25 DONE   dm.filerequests
2026-10-24 18:48:25 COMPLETED DM maintenance: 1 table(s). Log: logs/dm_maintenance_20261024_184825.log
```

S03. Table without a script is refused, nothing runs: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	./run_maintenance.sh dm.no_such_table
Expected	'ERROR: no script tables/dm.no_such_table.sql' before the connection prompt; nothing changed; exit code 1.
Checked	message shown | filerequests | files_active | exit code
Expected values	`yes|5000|2000|1`
Actual values	`yes|5000|2000|1`
Exit code	1

```
ERROR: no script tables/dm.no_such_table.sql for table dm.no_such_table - nothing was run
```

S04. Connection error is reported, nothing runs: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	Connection prompt answered with database no_such_db.
Expected	'ERROR: cannot connect', nothing changed, exit code 1.
Checked	message shown | filerequests | exit code
Expected values	`yes|5000|1`
Actual values	`yes|5000|1`
Exit code	1

```
2026-10-24 18:48:26 ERROR: cannot connect to postgres@localhost:55432/no_such_db (details in
    logs/dm_maintenance_20261024_184826.log)
```

S05. Commented-out table is not touched: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	Copy of run_maintenance.sh with # in front of dm.filerequests in TABLES.
Expected	dm.files done; dm.filerequests never started and still 5,000 rows; 'COMPLETED: 1 table(s)'.
Checked	files_active | filerequests | filerequests started | COMPLETED 1 table
Expected values	`0|5000|no|yes`
Actual values	`0|5000|no|yes`
Exit code	0

```
2026-10-24 18:48:26 START DM maintenance on postgres@localhost:55432/testdb: dm.files
2026-10-24 18:48:26 START  dm.files
dm.files done: partition dm.files_active emptied
2026-10-24 18:48:26 DONE   dm.files
2026-10-24 18:48:26 COMPLETED DM maintenance: 1 table(s). Log: logs/dm_maintenance_20261024_184826.log
```

S06. Partition missing: run stops, next table not run: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5. Then partition files_active is dropped.
How it is run	./run_maintenance.sh
Expected	'relation "dm.files_active" does not exist', dm.files FAILED, dm.filerequests never started, exit code 1.
Checked	message shown | FAILED logged | next table started | filerequests | exit code
Expected values	`yes|yes|no|5000|1`
Actual values	`yes|yes|no|5000|1`
Exit code	1

```
2026-10-24 18:48:26 START DM maintenance on postgres@localhost:55432/testdb: dm.files dm.filerequests
2026-10-24 18:48:26 START  dm.files
ERROR:  relation "dm.files_active" does not exist
2026-10-24 18:48:26 FAILED dm.files - maintenance stopped, the tables after it were not run (error above)
```

S07. Hard-coded partition: exactly the named partition is emptied: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	Copy of dm.files.sql with the TRUNCATE line changed to files_removed.
Expected	Exactly the named partition (files_removed, 3,000 'Y' rows) is emptied, with no check; files_active and files_other untouched; exit code 0.
Checked	files_removed | files_active | files_other | exit code
Expected values	`0|2000|0|0`
Actual values	`0|2000|0|0`
Exit code	0

```
2026-10-24 18:48:26 START DM maintenance on postgres@localhost:55432/testdb: dm.files
2026-10-24 18:48:26 START  dm.files
dm.files done: partition dm.files_removed emptied
2026-10-24 18:48:26 DONE   dm.files
2026-10-24 18:48:26 COMPLETED DM maintenance: 1 table(s). Log: logs/dm_maintenance_20261024_184826.log
```

S08. Foreign key: TRUNCATE fails, table rolled back: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5. Plus dm.child with a foreign key to dm.filerequests.
How it is run	./run_maintenance.sh dm.filerequests
Expected	TRUNCATE refused, everything undone: 5,000 rows, no copy left, exit code 1.
Checked	message shown | filerequests | copy table | exit code
Expected values	`yes|5000|no backup|1`
Actual values	`yes|5000|no backup|1`
Exit code	1

```
2026-10-24 18:48:26 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:26 START  dm.filerequests
copy OK: 2000 rows
ERROR:  cannot truncate a table referenced in a foreign key constraint
2026-10-24 18:48:26 FAILED dm.filerequests - maintenance stopped, the tables after it were not run (error above)
```

S09. Keep condition matching 0 rows is refused: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	Copy of dm.filerequests.sql with WHERE statusid IN (99).
Expected	'Copy check failed', table unchanged, no copy left, exit code 1.
Checked	message shown | filerequests | copy table | exit code
Expected values	`yes|5000|no backup|1`
Actual values	`yes|5000|no backup|1`
Exit code	1

```
2026-10-24 18:48:27 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:27 START  dm.filerequests
ERROR:  Copy check failed - nothing was changed
2026-10-24 18:48:27 FAILED dm.filerequests - maintenance stopped, the tables after it were not run (error above)
```

S10. Invalid keep condition is rolled back: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	Copy of dm.filerequests.sql with WHERE no_such_column = 1.
Expected	Error 'column does not exist', table unchanged, no copy left, exit code 1.
Checked	message shown | filerequests | copy table | exit code
Expected values	`yes|5000|no backup|1`
Actual values	`yes|5000|no backup|1`
Exit code	1

```
2026-10-24 18:48:27 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:27 START  dm.filerequests
ERROR:  column "no_such_column" does not exist
2026-10-24 18:48:27 FAILED dm.filerequests - maintenance stopped, the tables after it were not run (error above)
```

S11. Second run on the same day fails safely: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5.
How it is run	./run_maintenance.sh dm.filerequests, twice
Expected	Second run stops: today's maintenance table already exists; table and copy keep 2,000 rows; exit code 1.
Checked	message shown | filerequests | rows in copy | exit code
Expected values	`yes|2000|2000|1`
Actual values	`yes|2000|2000|1`
Exit code	1

```
2026-10-24 18:48:27 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:27 START  dm.filerequests
ERROR:  relation "maintenance_filerequests_20261024" already exists
2026-10-24 18:48:27 FAILED dm.filerequests - maintenance stopped, the tables after it were not run (error above)
```

S12. No count(*) on the main table: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5. PostgreSQL's full-read counter (pg_stat_user_tables.seq_scan) is read before and after.
How it is run	./run_maintenance.sh dm.filerequests
Expected	Counter +3: 1 = the copy step, 2 = TRUNCATE rebuilding the 2 indexes on the empty table. No count(*) on the table.
Checked	full reads | filerequests
Expected values	`3|2000`
Actual values	`3|2000`
Exit code	0

```
2026-10-24 18:48:29 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:29 START  dm.filerequests
copy OK: 2000 rows
check OK: 2000 rows copied back
dm.filerequests done: kept 2000 rows, removed about 3000 rows, copy in dm.maintenance_filerequests_20261024
2026-10-24 18:48:29 DONE   dm.filerequests
2026-10-24 18:48:29 COMPLETED DM maintenance: 1 table(s). Log: logs/dm_maintenance_20261024_184829.log
```

S13. Table in use: lock timeout after 60 s: PASS

Item	Detail
Dummy data	reset_data.sql: dm.files 5,000 rows LIST partitioned by is_removed (files_active 'N' 2,000, files_removed 'Y' 3,000); dm.filerequests 5,000 rows, statusid 1-5. Another session holds a lock on dm.filerequests for 90 seconds.
How it is run	./run_maintenance.sh dm.filerequests
Expected	After 60 s 'canceling statement due to lock timeout'; nothing changed, no copy left, exit code 1.
Checked	message shown | filerequests | copy table | exit code
Expected values	`yes|5000|no backup|1`
Actual values	`yes|5000|no backup|1`
Exit code	1

```
2026-10-24 18:48:35 START DM maintenance on postgres@localhost:55432/testdb: dm.filerequests
2026-10-24 18:48:35 START  dm.filerequests
ERROR:  canceling statement due to lock timeout
2026-10-24 18:49:35 FAILED dm.filerequests - maintenance stopped, the tables after it were not run (error above)
```

Observations

- **Exit codes:** `run_maintenance.sh` ends with 0 when every table is done and 1 when a table fails; the failing table and the error are in the log.
- **The tests never change the real scripts:** each scenario runs a copy of `run_maintenance.sh` and `tables/` in `tests/results/`, changed only where the scenario says.
- **Where:** this run used a local Docker test container (PostgreSQL 16.15). The Ubuntu test server was not reachable at the time; rerun the same tests there with `PGHOST=<server> ./tests/run_tests.sh` when it is back.

Not covered

- **Production-size data:** these runs use 5,000 rows. Timing for the outage window must be measured on a production-size copy.
- **Statement timeout (55 min):** set on every run but not triggered in these tests.
- **Generated or GENERATED ALWAYS identity columns:** with `SELECT *` such a table fails at the copy-back step and is rolled back; not in the current table list.
- **Production partition names:** the partition to empty is hard-coded (`dm.files_active` for 'N') and not checked. Check the name against production before the run; `\d+ dm.files` prints every partition and its rule in the log.